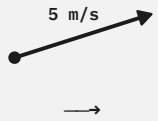
Picture dictionary: vectors

Language goal: I can **explain** how to add arrows.



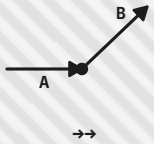
vector

vector · vetor

size + direction

my language: _____

use it: _____



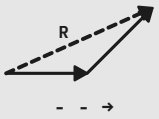
tip-to-tail

punta con cola · ponta com cauda

start the next arrow at the tip

my language: _____

use it: _____



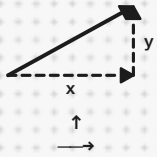
resultant

resultante · resultante

one arrow for the whole trip

my language: _____

use it: _____



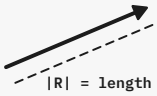
component

componente · componente

one part of an arrow

my language: _____

use it: _____



|R|

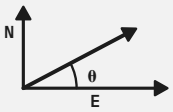
magnitude

magnitud · magnitude

the size of an arrow

my language: _____

use it: _____



∠ from E

north of east

al norte del este · ao norte do leste

angle from east, turning north

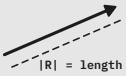
my language: _____

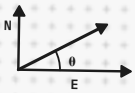
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

S2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

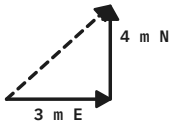
UNIT 1 · DAY 6 · S2 · SEI · PARTNER TALK

Talk it through: vectors

Language goal: I can **explain** how to add arrows.

DIRECTIONS · Read your lines aloud. Listen, then write. Switch roles each round.

ROUND 1 · 3 m EAST, THEN 4 m NORTH

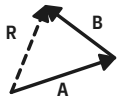


A ASKS "How far is the end from the start?"

B SAYS "I draw the arrows tip to ____. The result is √(____ + ____) = ____ m."

A CHECKS "So you're saying it's not 7 m because ____."

ROUND 2 · SWITCH · A = (+6, +2), B = (−4, +3)

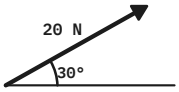


B ASKS "What is A + B?"

A SAYS "I add the east parts: ____. I add the north parts: ____. So R = (____ , ____)."

B ADDS "I want to add that we keep each ____ in its own column."

ROUND 3 · BOTH · 20 N AT 30° ABOVE +x



A "The x part is next to the angle, so I use ____: F_x = ____ N."

B "The y part is across from the angle, so I use ____: F_y = ____ N."

LISTENING LOG · MY PARTNER SAID…

NOW I WRITE IT MYSELF

To add two vectors, I…

Stems: I agree because… · Can you say that again? · What does ___ mean? · I want to add…

S2 (1) (2) (3)

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

The treasure heist

06 T

A thief's phone logged each move. Reconstruct where the treasure went.

EXHIBIT A · MOVE LOG (east = +x, north = +y)

MOVE	EAST–WEST	NORTH–SOUTH
1	+6 m	+2 m
2	−4 m	+3 m
3	+5 m	0 m
TOTAL		

ANALYSIS 1 · RESULT SIZE

$|R| = \sqrt{(R_x^2 + R_y^2)}$

ANALYSIS 2 · DIRECTION

$\theta = \tan^{-1}(R_y \div R_x)$ north of east

EXHIBIT B · DRAW THE ROUTE TIP-TO-TAIL (1 square = 1 m)

WITNESS · TRUE OR FALSE?

"The thief walked 6 + 2 + 4 + 3 + 5 = 20 m, so the treasure is 20 m away."

THE VERDICT

The treasure is ____ m from the start at ____° north of east, because ____

S2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Vector and component

06 H

DIRECTIONS · Fill all four boxes for each term. Sketch in at least one.

Vector

DEFINITION · MY WORDS

HOW TO ADD

tip-to-____ · add each ____

EXAMPLES

NON-EXAMPLES (scalars)

Component

DEFINITION · MY WORDS

FORMULAS

$V \cos\theta \cdot V \sin\theta$

SKETCH AN EXAMPLE

COMMON MIX-UP

One sentence using both terms: _____

S2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Four steps, every vector

06 M

DIRECTIONS · Work each problem through all four boxes.

PROBLEM 1 · A = (+6, +2) m and B = (−4, +3) m. Find R, |R|, and its angle.

1 READ

A_x ____ A_y ____ · B_x ____ B_y ____

2 DRAW

tip-to-tail sketch

3 SOLVE

R_x = A_x + B_x · |R| = √(R_x² + R_y²) · θ = $\tan^{-1}$(R_y/R_x)

4 WRITE

R is ____ m at ____° north of east.

PROBLEM 2 · A 20 N pull points 30° east of north. Let east be +x. Find F_x and F_y.

1 READ

F = ____ · angle measured from ____

2 DRAW

draw it: which part is next to θ?

3 SOLVE

F_x = 20 sin 30° · F_y = 20 cos 30°

4 WRITE

F_x = ____ N and F_y = ____ N. I used sin for x because ____.

Hardest box: □ Read □ Draw □ Solve □ Write

S2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

I have... who has? Vectors

06 G

DIRECTIONS · Start at ★. Solve, find the tile that "has" your answer, and keep going.

★ START — **A**

WHO HAS…

4 m east + 2 m west. Result?

I have **68.2°** — **B**

WHO HAS…

20 N at 30° above +x. F_x = 20 cos 30°?

I have **(+2, +5)** — **C**

WHO HAS…

|R| for (+2, +5)? (√(4 + 25))

I have **10.0 N** — **D**

WHO HAS…

3 m east + 4 m north. Size of the result?

I have **2 m east** — **E**

WHO HAS…

A = (+6, +2), B = (−4, +3). R = A + B?

I have **5 m** — **F**

WHO HAS…

Is 3 east + 4 north a 7 m diagonal?

I have **5.39** — **G**

WHO HAS…

$\theta = \tan^{-1}(5 \div 2)$, north of east?

I have **17.3 N** — **H**

WHO HAS…

Same pull. F_y = 20 sin 30°?

I have **no − 5 m** — **I**

END

MY PATH: A → ___ → ___ → ___ → ___ → ___ → ___ → ___ → END

S2 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Comic: why 3 + 4 can equal 5

06 CS

YOUR READER

A younger student who thinks two walks always add up the simple way.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ BLUE — Tip-to-tail arrows for the 3 m and 4 m legs
- ☐ ◌ BLUE — A dashed resultant arrow labeled **5 m**
- ☐ ◯ PENCIL — The angle θ and the words "north of east"

1 · WALK EAST *A character walks 3 m east.*

CAPTION _____

2 · WALK NORTH *Then walks 4 m north.*

CAPTION _____

3 · THE SHORTCUT *A friend walks straight from start to finish.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **vector · resultant · tip-to-tail · magnitude**

S2 1 2 3

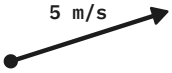
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · resultant

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Resultant · resultante (ES) · resultante (PT)



TODAY'S CASE

Walk 3 m east, then 4 m north. One arrow can show your change in position.

01 · SKETCH AND LABEL

On a grid draw both legs tip-to-tail. Give every square the same scale.

02 · CHANGE THE PICTURE

Draw the blue start-to-finish arrow. Compare its length with the total distance walked.

Annotate your sketches. Circle the mark that best shows resultant. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

The single arrow represents …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else